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Why Do Irrelevant Alternatives Matter? An fMRI-TMS Study of Context-Dependent Preferences

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Abstract

Both humans and animals are known to exhibit a violation of rationality known as "decoy effect": introducing an irrelevant option (a decoy) can influence choices among other (relevant) options. Exactly how and why decoys trigger this effect is not known. It may be an example of fast heuristic decision-making which is adaptive in natural environments, but may lead to biased choices in certain markets or experiments. We used functional magnetic resonance imaging (fMRI) and transcranial magnetic stimulation (TMS) to investigate the neural underpinning of the decoy effect of both sexes. The left ventral striatum was more active when the chosen option dominated the decoy. This is consistent with the hypothesis that the presence of a decoy option influences the valuation of other options, making valuation context-dependent even when choices appear fully rational. Consistent with the idea that control is recruited to prevent heuristics from producing biased choices, the right inferior frontal gyrus (rIFG), often implicated in inhibiting prepotent responses, connected more strongly with the striatum when subjects successfully overrode the decoy effect and made unbiased choices. This is further supported by our TMS experiment: subjects whose rIFG was temporarily disrupted made biased choices more often than a control group. Our results suggest that the neural basis of the decoy effect could be the context-dependent activation of the valuation area. But the differential connectivity from the frontal area may indicate how deliberate control monitors and corrects errors and biases in decision-making.

Significance Statement

Standard theories of rational decision-making assume context-independent valuations of available options. Motivated by the importance of this basic assumption, we used functional magnetic resonance imaging to study how the human brain assigns values to available options. We found activity in the valuation area to be consistent with the hypothesis that values depend on irrelevant aspects of the environment, even for subjects whose choices appear

42 fully rational. Such context-dependent valuations may lead to biased
43 decision-making. We further found differential connectivity from the
44 frontal area to the valuation area depending on whether biases were
45 successfully overcome. This suggests a mechanism for making ratio-
46 nal choices despite the potential bias. Further support was obtained
47 by a transcranial magnetic stimulation experiment, where subjects
48 whose frontal control was temporarily disrupted made biased choices
49 more often than a control group.

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56 the manuscript.

57 1 Introduction

58 In standard rational choice theory, decision-making is represented as assigning
 59 value to each feasible option and choosing the option with the highest value.
 60 Moreover, the value of an option should depend only on its intrinsic properties,
 61 not on the context in which it appears. Neuroscientific studies suggest that val-
 62 ues of available options are indeed encoded (Padoa-Schioppa and Assad, 2006),
 63 but there is mixed evidence on the extent to which these values depend on the
 64 context (Nieuwenhuis et al., 2005; Padoa-Schioppa and Assad, 2008). In this
 65 article, we use the well-known *decoy effect* to study context-dependence in the
 66 human brain.

67 A decoy is an option which is unlikely to be chosen but may influence choice
 68 among other options. To illustrate, consider *The Economist* magazine’s adver-
 69 tisement of three options: print-only and print-and-web subscriptions, each at
 70 125 dollars per year, and web-only subscription at 59 dollars per year (Ariely,
 71 2008). The print-only subscription was a decoy, because it was *dominated* by
 72 print-and-web: it is clearly worse to get only print access than to get both print
 73 and web accesses at the same price. The domination was *asymmetric* in the
 74 sense that the decoy (print-only) was dominated only by one option, called the
 75 *target* (print-and-web), and not by the other, called the *competitor* (web-only).
 76 In the marketing literature, the target is a promoted product, the competitor
 77 a similar product not currently being promoted, and the decoy introduced to
 78 boost the demand for the target. Though consumers are not expected to choose
 79 the decoy, they might be impelled by the asymmetric domination to choose the
 80 target which could be favored because of the decoy instead of the competitor
 81 which might be penalized.

82 The marketing strategy contradicts standard rational choice theory, accord-
 83 ing to which adding an “irrelevant alternative” (the decoy) to the menu should
 84 not influence the values assigned to the target and the competitor. However,

85 across taxa (such as humans, gray jays, rufous hummingbirds, ants, and even
86 an amoeboid organism), the existence of a decoy does increase the frequency of
87 choice of the target - there is a decoy effect on choice behavior (Huber et al.,
88 1982; Shafir et al., 2002; Bateson et al., 2003; Sasaki and Pratt, 2011; Latty and
89 Beekman, 2011). Despite of its importance and prevalence in choice, the un-
90 derlying neural basis for the decoy effect is poorly understood. Maintaining the
91 assumption that decision-makers choose the option with the highest value, we
92 will investigate the context-dependence of values. Specifically, we hypothesize
93 that if the context (the menu from which options are chosen) includes a decoy
94 option, then the valuation of the target is increased relative to the competitor.
95 Such context-dependent valuations will necessarily deviate from the intrinsic
96 values of the options, which raises the issue of how the brain can avoid biased
97 choices.

98 We report two sets of experiments in which subjects made choices from
99 menus that included asymmetrically dominated decoys. In the fMRI experi-
100 ment, the first question we address is whether the value of the chosen option
101 is influenced by the decoy. Such context-dependent valuation could potentially
102 bias choices towards the target, causing a decoy effect on choice behavior. This
103 raises the next question of how the brain can correct the potential bias. In
104 addressing this question, we constructed a trial-by-trial measure of the strength
105 of the decoy effect and explored potential control modulation from the frontal
106 area for reducing the bias. Since the fMRI data only establish correlations, in
107 the TMS experiment we disrupted the frontal area found in the fMRI experi-
108 ment to examine whether the decoy effect on choice behavior becomes more
109 pronounced when control is weakened. Our results suggest that observable
110 context-dependent choices or decoy effects may be shaped by the interaction of
111 two processes: an evaluation process which is inherently context-dependent and
112 a control system which can possibly override context-dependence.

113 2 Materials and Methods

114 There were two experiments, the fMRI experiment and the TMS experiment.
115 We describe them in turn.

116 The fMRI Experiment

117 Subjects

118 Thirty-two right-handed healthy individuals (sixteen females; mean age 23.94)
119 participated in the fMRI experiment approved by the Institutional Review
120 Board of National Yang-Ming University. Each participated in two pre-tests,
121 the scanning experiment and a post-test. One trial from the scanning experi-
122 ment was randomly drawn when a subject left the scanner and she received a
123 voucher and cash payment corresponding to her choice, in total worth of 1500
124 New Taiwan dollars (approximately 50 US dollars). Each subject was also paid
125 300 dollars for each of the pre-tests and the post-test, totaling 900 dollars.

126 Task

127 Subjects made choices from a series of two- and three-option menus. Each
128 option was a restaurant meal with a specified price and a quality rating. The
129 price was the cost of a typical meal at the restaurant. The price and the rating
130 (with a higher percentage implying a better quality) were based on informa-
131 tion from the Taiwanese gourmet website *enjoygourmet.com*. In each trial, the
132 subject was endowed with 1500 dollars, from which the price of the meal she
133 selects would be deducted. Four trials are illustrated in Figure 1A. For instance,
134 the menu in the top left panel has two options. The left option is a meal at a
135 restaurant with a 72% rating, at a price of 680 dollars. If she chooses this option
136 she will be given a voucher for a meal at a restaurant with the specified quality
137 rating. She will receive what remains of her endowment, $1500 - 680 = 820$
138 dollars, in cash. The right option is a meal at a better restaurant; it has a 95%

rating and costs 1010 dollars. Thus, the trade-off facing the subject is that she can spend more of her endowment to get a better meal. In some trials, one of the options on the menu was an asymmetrically dominated decoy. By definition, such a decoy is dominated in every dimension, in the sense that it is both more expensive and has a lower quality than the target. For instance, in the bottom left panel, the middle option is the decoy and the left-most option is the target. The right-most option is the competitor, which does not dominate the decoy. In the marketing literature, there are many types of decoys. Compromise decoys are also a common type. The idea is to include an undominated yet very unattractive decoy to make the target a compromise and hence chosen. However, since compromise decoys are not dominated and can be chosen with substantial probability, the irrelevance of them is more questionable.

Experimental Design and Statistical Analysis of the fMRI Experiment

The experiment consists of three stages, the pre-test, the post-test, and the scanning experiment. Each pre-test and post-test consisted of 192 trials with two available options. They served to identify how each subject traded off price versus quality without the influence of decoys. The measured trade-off from the pre-tests was used to tailor-make each subject's decision problems in the scanning experiment. The post-test was administered right after the scanning experiment. The measured trade-off from the post-test was used to estimate each subject's utility function as explained in (A) below.

In the scanning experiment, the main focus was on *decoy trials* with three available options: two options A and B , neither dominating the other, and a third decoy option. The decoy trials were paired so that each undominated option would be the target in one trial and the competitor in the other. Such a *matching pair of decoy trials* has the form $\{A, B, A'\}$ and $\{A, B, B'\}$. In $\{A, B, A'\}$ the decoy A' is dominated by A (the target) but not by B (the

competitor), while in $\{A, B, B'\}$ the decoy B' is dominated by B (the target) but not by A (the competitor). The choice is said to be *along* the decoy if the target is chosen (say, A is chosen from $\{A, B, A'\}$) and *against* the decoy if the competitor is chosen (say, A is chosen from $\{A, B, B'\}$).

There were 192 trials in the scanning experiment. Each pair of options A and B appeared six times: once in a two-option choice problem $\{A, B\}$; twice in a three-option choice problem $\{A, B, A'\}$; twice in a three-option choice problem $\{A, B, B'\}$; once in a three-option choice problem $\{A, B, E\}$ where E was an undominated third option. By including $\{A, B, E\}$ trials with three undominated options, we avoided a situation where all three-option decision problems had an asymmetrically dominated option, which would have been too repetitive and predictable. The order of the trials and the ordering of options in a trial were randomized, with one exception: for every $\{A, B, A'\}$ trial there was a matching $\{A, B, B'\}$ trial in which the ordering of the options on the screen (in terms of target, competitor and decoy) was controlled to be the same, so that the only difference between these two trials was that the decoy changed and the roles of A and B (in terms of being target or competitor) were reversed (as illustrated in the two bottom panels of Figure 1A). Note that the similarity of the two matching trials could potentially reduce the decoy effect.

The scanning experiment was self-paced. In each trial, pictures of foods (to remind subjects the experiment involved restaurant meals) were presented for 0.5 second, after which a fixation cross (randomized between 0 to 2 seconds) appeared. The choice screen followed and was on until the subject made her choice. Finally, the fixation cross appeared again during the jittered inter-trial interval (randomized between 2 to 6 seconds) before the next trial began.

For each subject, we obtained contrasts we are interested in. Parameter estimates from contrasts in single subject models were entered into the second level random-effect analysis to carry out a one-sample t-test on whether the activation of a contrast was significantly different from zero after whole-brain cluster

correction. The criterion we used in all analyses is cluster-defining threshold uncorrected p-value 0.001 and familywise error rate 0.05. We also verified our key results are robust using non-parametric cluster-level inference (Eklund et al., 2016) (data not shown). In all general linear models (GLMs) and psychophysiological interaction (PPI) analysis 1 indicated below, the degree of freedom is 29. In PPI 2, it is 28.

fMRI Data Acquisition and Preprocessing

Imaging was performed using a 3T Siemens Trio MR scanner. An EPI sequence was used to acquire the functional data (time repetition = 2000 ms, time echo = 30 ms, flip angle = 90° , voxel size = 3.4mm x 3.4mm x 3.4mm). The functional (T2* weighted, BOLD) images were collected in 33 transverse planes parallel to the anterior commissure-posterior commissure line to cover the whole brain. There were four EPI sessions so that each session had 192/4 = 48 trials. A high-resolution T1 weighted MP-RAGE anatomical MRI volume scan of the entire head was run once on each subject (time repetition = 2530 ms, time echo = 3.03 ms, flip angle = 7° , voxel size = 1mm x 1mm x 1mm).

Data analysis was performed using SPM8. The functional images were corrected for differences in slice-acquisition time, realigned, coregistered, normalized to the MNI average template and smoothed. The first four images were discarded from the analysis for magnetization to reach steady state. Two subjects were removed due to excessive head motions.

fMRI Data Analysis

The fMRI analysis has two parts. Part (A) studies decoy-induced changes in valuations. Part (B) addresses how these changes might be overcome to restore rational (unbiased) choice.

(A) Context-Dependent Values

222 We first addressed the hypothesis that the valuation of the chosen option
 223 is influenced by the decoy. We estimated a linear utility function from choices
 224 in the post-test with only two available options. In the absence of decoys, this
 225 utility function reflects the intrinsic values of the options. We then constructed
 226 a comprehensive model where various value signals and an indicator of Along
 227 versus Against trials are included. Our aim was to find regions correlating with
 228 valuations and to see how the value signals differentiate between Along and
 229 Against trials.

230 To estimate a linear utility function, consider a trial $\{A, B\}$ in the post-test
 231 where the price of A (resp. B) is P_A (resp. P_B) and the quality is Q_A (resp.
 232 Q_B). The intrinsic utility of choosing A is $u(A) = b_P(1500 - P_A) + b_Q Q_A$, where
 233 b_P and b_Q are weights put on the cash payment $1500 - P_A$ and the quality Q_A .
 234 This is called intrinsic utility since it only depends on the price and quality of
 235 the option. The intrinsic utility of B is $u(B) = b_P(1500 - P_B) + b_Q Q_B$. Using
 236 the logit function, the probability of choosing A is

$$\frac{1}{1 + \exp^{-\lambda(u(A) - u(B))}}$$

237 where λ is the precision parameter. We normalized so that $b_P + b_Q = 1$, and
 238 estimated b_P and λ by maximum likelihood. We then used the estimated b_P
 239 to calculate the intrinsic utility of each option for each trial of the scanning
 240 experiment.

241 We ran GLM 1 with three regressors of interest. They are convolved with
 242 the canonical form of the hemodynamic function and will not be mentioned later
 243 on. The first regressor is to model the half-second when subjects saw pictures of
 244 foods (called picture). The second is to model the motor response when subjects
 245 made choices (called motor). The third is to model the entire duration of the
 246 choice. To account for the values of each option and whether choice is Along
 247 or Against in every trial, we included six parametric modulations for the third
 248 regressor.

249 We attempted to control for those value signals that would most likely be
 250 behaviorally relevant. The value of the chosen option is certainly important
 251 and is defined for all trials. Undominated unchosen options may also influence
 252 the decision process. For two-option trials and decoy trials, we control for the
 253 value of the unique undominated unchosen option. In $\{A, B, E\}$ trials there
 254 are actually two undominated unchosen options; the one with the highest value
 255 is the one most likely to influence the decision process by competing with the
 256 chosen option, but we cannot rule out that the worst unchosen option also plays
 257 a role. Therefore, for $\{A, B, E\}$ trials we separately include the values of both
 258 the best and the worst unchosen options. Finally, the value of the decoy is also
 259 controlled for.

260 Specifically, the six parametric modulations are as follows. The first is on
 261 the utility of the chosen option (defined for all trials). The second is on the
 262 utility of the undominated unchosen option in two-option trials and decoy trials
 263 (taking the value of 0 for $\{A, B, E\}$ trials). The third is on the utility of the
 264 highest-valued unchosen option in $\{A, B, E\}$ trials (taking the value of 0 for
 265 two-option and decoy trials). The fourth is on the utility of the decoy in decoy
 266 trials (taking the value of 0 for two-option and $\{A, B, E\}$ trials). The fifth is on
 267 the utility of the lowest-valued unchosen option in $\{A, B, E\}$ trials (taking the
 268 value of 0 for two-option and decoy trials). The sixth parametric modulation
 269 is an indicator of Along versus Against (taking the value of 1 when choice is
 270 along the decoy, -1 when choice is against the decoy, and 0 for two-option and
 271 $\{A, B, E\}$ trials). Note that we shut down the orthogonalization procedure in
 272 SPM. We thus made it harder to obtain significance, since the un-orthogonalized
 273 value modulations are correlated.

274 Lastly, the six head motion parameters are included as the regressors of
 275 no interest. They are always included in all GLMs and will not be mentioned
 276 later on. The contrast of the first parametric modulation allows us to find
 277 brain correlates of the utility of the chosen option, other utility-related mea-
 278 sures being controlled for. Similarly, the sixth modulation allows us to test

279 which brain region is more active when choice is along (rather than against) the
 280 decoy, again controlling for the possible influence of the utility-related measures.
 281 This amounts to a simple additive effect of the context. For between-subject
 282 correlation of striatal Along-Against activity and two behavioral measures of
 283 how “decoyable” a subject is, we extract the coefficients of the Along-Against
 284 contrast within a 4-mm sphere at three peak voxels of putamen and one peak
 285 voxel of caudate.

286 (B) Overcoming the Decoy Effect

287 The results in (A) are consistent with the hypothesis that context-dependent
 288 values deviate from intrinsic utilities. This may bias choices in favor of the tar-
 289 get, which raises the question of whether the brain has some mechanism for
 290 correcting the potential bias. We hypothesized that regions related to this
 291 mechanism would, on the one hand, correlate negatively with the strength of
 292 the decoy effect and on the other hand, would differentially connect with the
 293 valuation area depending on whether or not the decoy effect was successfully
 294 overridden. We constructed a trial-by-trial measure of the strength of the decoy
 295 effect $d(t)$ (see (1) below) and identified brain regions correlating negatively with
 296 it. If these regions are part of a mechanism for mitigating a decoy-induced bias,
 297 then since (A) suggested that decoy-induced changes in valuations may be re-
 298 flected in differential striatal activity, the regions identified should differentially
 299 connect with the striatum depending on whether the decoy effect is successfully
 300 overridden.

301 To investigate which brain areas correlate with $d(t)$, we ran GLMs 2 and
 302 3. In GLM 2, for each subject, we classified the decoy trials into two halves
 303 according to the size of $d(t)$. The half with a stronger decoy effect (a larger
 304 $d(t)$) was classified as Strong. The half with a weaker decoy effect (a smaller
 305 $d(t)$) was classified as Weak. We include six regressors of interest. The binary
 306 classification gives Strong and Weak as two regressors. The next two are on trials
 307 with only two options to choose from (classified as two-option trials) and on trials

308 with three undominated options (classified as three-undominated-option trials).
 309 These four regressors model the entire duration of the choice. Lastly, picture
 310 and motor regressors are added. The final four regressors (the first modeling
 311 two-option trials, the second modeling trials with three undominated options,
 312 the picture and the motor regressors) will be called the four standard control
 313 regressors.

314 To find regions coding $d(t)$ parametrically we tried GLM 3. There are six
 315 regressors of interest. The first is a regressor modeling the decoy trials. For
 316 this regressor, we included a parametric modulation $d(t)$. These two regressors
 317 model the entire duration of the choice. We also included the four standard
 318 control regressors.

319 The binary classification (Weak versus Strong) in GLM 2 is coarser and
 320 depends less on the functional form of (1) than the parametric modulation
 321 in GLM 3. We identified the inferior parietal lobule (IPL) in both. We also
 322 performed several robustness checks on IPL's negative correlation with $d(t)$,
 323 including allowing for the possibility that the decision bound T in (1) might
 324 have changed during the scanning experiment and measuring the strength of
 325 the decoy effect in two model-free ways. We also explored a model of random
 326 utility and found its neural correlates.

327 The fact that IPL negatively correlates with $d(t)$ suggests that it may play a
 328 role in a control mechanism for reducing the decoy effect. Because (A) indicated
 329 that the decoy effect is reflected in differential striatal activity, we explored how
 330 IPL might differentially connect with the striatum depending on whether the
 331 decoy effect is successfully reduced. We did not find a direct modulation from
 332 IPL in our data. We did find a plausible indirect modulation through the right
 333 inferior frontal gyrus (rIFG) which we will further verify by a TMS experiment.
 334 The indirect modulation is based on two PPIs.

335 PPI 1 between IPL and rIFG

336 A decoy trial $\{A, B, A'\}$ is a Conflict trial if $u(A) - u(B) < 0$ so the competi-
 337 tor is intrinsically better. It is a NoConflict trial if $u(A) - u(B) > 0$ so the target
 338 is intrinsically better. Exerting control to reduce the decoy effect is important
 339 in Conflict trials, in order to prevent the intrinsically inferior target from being
 340 chosen. In NoConflict trials, control is less important since the option favored
 341 by the decoy effect, the target, is anyway the intrinsically better option.

342 We hypothesized that when IPL is active and control is more important, the
 343 connectivity between IPL and a control area might be stronger. We took the
 344 average activity of a 4-mm sphere surrounding the IPL peak voxel (-40, -40, 46)
 345 identified in GLMs 2 and 3 as the seed. First, we estimated a GLM which was
 346 built solely for PPI 1. There are six regressors of interest. We have Conflict
 347 trials and NoConflict trials as two regressors. They model the entire duration
 348 of the choice. We also included the four standard control regressors. Second,
 349 the time series of the average activity of the 4-mm IPL sphere was extracted,
 350 removing the variances associated with the head motion parameters. Third, we
 351 ran PPI 1 with the psychophysiological interaction of the seed activity and the
 352 contrast of Conflict-NoConflict. The psychological contrast Conflict-NoConflict
 353 and the bold eigenvariate were also included as regressors.

354 PPI 2 between rIFG and Striatum

355 We hypothesized that in trials where control is important, the connectivity
 356 between control and valuation areas may be stronger when control is successful
 357 compared with when control is unsuccessful. As argued in PPI 1, control is
 358 important in Conflict trials, to make sure that the intrinsically best option (the
 359 competitor) is chosen against the decoy. Hence in Conflict trials, control is
 360 successful if the competitor is chosen, i.e., in Against trials. It is unsuccessful if
 361 the target is chosen, i.e., in Along trials.

362 To explore the hypothesis, we took the average activity of a 4-mm sphere
 363 surrounding the peak voxel (60, 16, 22) of rIFG identified in PPI 1 as the seed.

364 First, we estimated a GLM which was built solely for PPI 2. There are seven
 365 regressors of interest. There are mainly two regressors for Conflict trials: one for
 366 Against trials (SuccessfulControl) and one for Along trials (UnsuccessfulCon-
 367 trol). In addition, a regressor for NoConflict trials is added. These three regres-
 368 sors model the entire duration of the choice. We also included the four standard
 369 control regressors. Second, the time series of the average activity of the 4-mm
 370 rIFG sphere was extracted, removing the variances associated with the head mo-
 371 tion parameters. Third, we ran PPI 2 with the psychophysiological interaction
 372 of the seed activity and the contrast of SuccessfulControl-UnsuccessfulControl.
 373 The psychological contrast SuccessfulControl-UnsuccessfulControl and the bold
 374 eigenvariate were also included as regressors. The results were based on twenty-
 375 nine subjects, since one subject managed to always choose the intrinsically bet-
 376 ter option. Eight subjects always chose the intrinsically better options in some
 377 EPI session, so the contrast was scaled by the number of sessions in which it
 378 could be built. The scaled contrasts were then entered into the second level
 379 random-effect analysis.

380 The TMS Experiment

381 Subjects

382 Sixty-four healthy individuals (twenty-five females; mean age 22.70) partic-
 383 ipated in the TMS experiment approved by the Institutional Review Board of
 384 Taipei Veterans General Hospital. The TMS experiment closely resembled the
 385 fMRI experiment. Each subject participated in a pre-test, the TMS session and
 386 a post-test. One trial from the TMS session was randomly drawn, and the sub-
 387 ject received a voucher and cash payment corresponding to her choice, in total
 388 worth of 1500 New Taiwan dollars. Each subject was also paid 300 dollars for
 389 the pre-test and the post-test, totaling 600 dollars.

390 Task

391 The task is the same as that in the fMRI experiment. Subjects made choices
 392 from a series of two- and three-option menus.

393 Experimental Design and Statistical Analysis of the TMS Exper- 394 iment

395 The pre-tests and post-tests are the same as those for the fMRI experiment.
 396 The post-tests were administered 1.5 hours after TMS was delivered, when
 397 the effect should have expired (Huang et al., 2005). The TMS session closely
 398 resembles the fMRI experiment except that all trials have three options. There
 399 were 216 trials. Among them, 192 trials are decoy trials in matching pairs. The
 400 remaining 24 trials are $\{A, B, E\}$ trials. The main focus of the analysis is on the
 401 matching pairs. Since the effect of TMS is estimated to be one hour at most,
 402 we did not include two-option trials to have as many decoy trials as possible.
 403 A few $\{A, B, E\}$ trials were included to avoid the task being too repetitive and
 404 predictable.

405 The TMS session was self-paced. In each trial, pictures of foods were pre-
 406 sented for 0.5 second, after which a fixation cross of 0.25 second appeared. The
 407 choice screen followed and was on until the subject made her choice. Finally, the
 408 fixation cross appeared again during the jittered inter-trial interval (randomized
 409 between 2 to 3 seconds) before the next trial began.

410 The degree of freedom for between-group t-test is 1. That for within-group
 411 t-test is 31.

412 TMS Parameters and Sites

413 We applied theta burst TMS before subjects started the choice task. The
 414 TMS pulses were applied in a continuous theta burst TMS pattern as specified
 415 by a previous study (Huang et al., 2005). Because this theta burst TMS proto-
 416 col can induce longer effects on neural activities, it is especially suitable for our
 417 current experimental design due to long response time (RT). Subjects received

3 pulses of stimulation at 50 Hz which were repeated every 200 milliseconds for 40 seconds at 40% of maximum output of 2.53 Tesla. This roughly corresponds to 47% to 73% of typical Taiwanese subject motor thresholds. We used a customized figure-of-8 flat coil of 55 mm and a Magstim Rapid2 Stimulator (Whitland, Wales, United Kingdom) to deliver the pulses. A fixed stimulation level was used because it has been proven successful and replicable in many studies and over a wide range of tasks (Kim et al., 2011). Moreover, motor cortex excitability may not provide a good guide to TMS thresholds in other cortical areas (Stewart et al., 2001).

According to the fMRI result, rIFG could be playing a role in implementing control to reduce decoy-induced decision-making bias. Yet since IPL functionally connects more strongly with rIFG in Conflict than in NoConflict trials, potentially, both rIFG and IPL could be our region of interest for the TMS experiment. In the literature, rIFG is often implicated in overriding irrelevant responses. Moreover, IFG and its vicinity have stronger anatomical and functional connections with the basal ganglion than IPL (Manes et al., 2014) and we argue that the striatum might be related to the decoy effect due to its differential Along versus Against activity, we therefore choose rIFG as the region of interest. If rIFG is temporarily disrupted, then the inhibitory control is expected to be weaker and the decoy effect is expected to be stronger.

The design is between-group. Subjects were randomly assigned to two groups so that each group has thirty-two subjects, the same number as we had for the fMRI experiment. In one group the site of stimulation was at the peak voxel of rIFG (60, 16, 22) identified from the fMRI experiment. The site was determined by each subject's T1-weighted anatomical image with Brainsight software (Rogue Research, Montreal, Canada). To determine the correspondence between the location on the skull and the site of stimulation on the T1 image, subjects wore goggles with a mounted Polaris infrared tracker (Northern Digital, Waterloo, Canada) attached to coregister them with their structural image. This helps pinpoint the skull point overlying the stimulated region to be iden-

448 tified. In the other group the site of stimulation was the vertex. The vertex is
 449 a point midway between the nasion and the inion and of equal distance to the
 450 right and left intertragal notches (Muggleton et al., 2008). It serves as a control
 451 site for the experiment. To make sure the procedures experienced by vertex
 452 subjects were exactly the same as for IFG subjects, we also used the Brainsight
 453 pen tool to point out (and localize) the vertex for stimulation. During stimula-
 454 tion over rIFG, the coil handle was oriented toward the superior portion of the
 455 head, whereas the coil handle was oriented toward the right side of the head
 456 during stimulation over the vertex.

457 3 Results

458 Behavior: Choices and RT

459 In terms of choices, 81.20% of the matching pairs of trials were behaviorally
 460 rational: the subject chose the same option from $\{A, B, A'\}$ and $\{A, B, B'\}$.
 461 16.41% exhibited preference reversals: the subject chose the target in both
 462 trials, i.e., she switched from A to B when the decoy changed from A' to B' .
 463 These numbers are in line with previous studies, which typically have preference
 464 reversals occurring about 20% of the time. Only 2.05% of the matching pairs
 465 exhibited anti-choices: the subject chose the competitor in both trials, i.e., she
 466 switched from B to A when the decoy changed from A' to B' . Thus, when
 467 choices were not behaviorally rational, there was a strong tendency to favor the
 468 target: A in $\{A, B, A'\}$ and B in $\{A, B, B'\}$. The subject-by-subject breakdown
 469 of the percentage of preference reversals and anti-choices are shown in Figure
 470 1B.

471 Since the great majority of matching pairs were behaviorally rational, *choices*
 472 in this experiment were largely context-independent, but this does not neces-
 473 sarily imply context-independent *valuations*. A decoy-induced increase in the

value of the target would not influence choice if the target has a higher intrinsic utility than the competitor. Even if the intrinsic utility of the target is lower, the decoy effect will not influence choice unless it is large enough to overcome the intrinsic utility difference that favors the competitor. However, neural activity can indicate changes in values that do not manifest themselves in choice behavior. The RT difference in behaviorally rational pairs provides the first evidence that values may depend on the context even when choices do not. Since a behaviorally rational pair consists of an Along trial (where the choice is along the decoy) and an Against trial (where the choice is against the decoy), if the same chosen option is valued more when it is the target than when it is the competitor, choosing this option will be easier in the Along trial than in the Against trial. Across all subjects, the average RT was 8.36 seconds in Along and 9.94 seconds in Against behaviorally rational trials, indicating that it was indeed easier to make a decision in the former situation. The average percentage decrease of RT in Along trials is 16.20%. The average percentage decrease of RT in Along than in Against behaviorally rational trials is strongly significantly different from zero over all 32 subjects, having the p-value (two-tailed) of 0.0000052 (Figure 1C).

Neural Signature of a Decoy Effect in the Valuation Area

We aim to find a neural manifestation of the decoy effect. If valuations are context-independent, then after controlling for the possible influence of various value signals, the valuation area should not differentiate between Along and Against trials. On the other hand if valuations are context-dependent, then options might be more highly valued when they dominate the decoy. Hence reward-sensitive areas are predicted to be more active in Along than in Against trials. This prediction was supported by our fMRI data. In GLM 1 we included various value signals and an indicator of Along or Against trials. We found that the striatum activity correlates parametrically with the estimated intrinsic utility of the chosen option positively (Figure 2A, Table 1). Also, the left ventral striatum was more active in Along than in Against trials (Figure 2B, Table 2).

504 The striatum region correlating with the value of the chosen option overlapped
 505 with that more active in Along than in Against trials (Figure 2C). That the same
 506 neural structure correlating with the value of the chosen option is more active
 507 when the decoy favors the chosen option is consistent with context-dependent
 508 valuations.

509 If the striatal differential activity Along versus Against indicates that val-
 510 uations are biased in favor of the target, then subjects whose striatal activity
 511 differentiates more strongly between Along and Against trials should exhibit a
 512 stronger decoy effect on choice behavior. Across subjects, we indeed find that
 513 the striatal Along versus Against activity is positively correlated with two be-
 514 havioral measures of how “decoyable” a subject is (Table 3). The first such
 515 measure is simply the number of preference reversals (Figure 2D). The second
 516 measure μ is the average strength of the decoy effect as explained below (Figure
 517 2E).

518 For completeness, we also report regions correlating with the utility differ-
 519 ence of the chosen option and the undominated unchosen option; the latter
 520 competes with the chosen option and presumably influences decision-making
 521 (Table 4). Further, we find no brain region correlating with the utility sum of
 522 the chosen option and the undominated unchosen option.

523 Trial-by-trial Strength of the Decoy Effect

524 How well an option will satisfy a person’s wants and needs depends on the
 525 intrinsic properties of price and quality. If valuations are influenced by irrelevant
 526 aspects of the context, as suggested by the higher striatal activity in Along than
 527 in Against trials, choices may become biased towards the target and an intrinsi-
 528 cally worse option may be chosen. Unbiased choice may therefore require some
 529 cognitive control. To find out how cognitive control could be recruited and its
 530 potential effect on valuation, we first constructed a measure of the trial-by-trial
 531 strength of the decoy effect using a simplified linear ballistic accumulator (LBA)

model (Brown and Heathcote, 2008). Note that the number of preference reversals of a subject can be used to compare between-subject who is more decoyable, but cannot be used to determine *within-subject* when she experiences a stronger decoy effect. For this, we need a trial-by-trial measure. We hypothesized that regions related to a putative control mechanism will correlate negatively with the trial-by-trial strength of the decoy effect. Moreover, if the decoy-induced change in valuation manifest itself in differential striatal activity, regions related to control may exhibit differential connectivity with the striatum depending on whether the bias induced by decoys is successfully overridden. We will explore these hypotheses after discussing the LBA model and the trial-by-trial measure.

The idea behind the LBA model is that the time to make a decision is inversely proportional to the “evidence” in favor of the decision. In a decoy trial, since the decoy is dominated and may no longer be considered relevant as decision goes, the relevant objects of choice are the chosen option and the undominated unchosen option. The *intrinsic* utility difference between them is $u_{chosen} - u_{unchosen}$, where the intrinsic utilities u_{chosen} and $u_{unchosen}$ were estimated in (A). We hypothesized that the decoy effect in trial t causes the relative value assigned to the target to be shifted by some amount $d(t)$. We do not impose a sign on $d(t)$ although in general we would expect $d(t)$ to be positive. The strength of relative evidence favoring the chosen option will be $u_{chosen} - u_{unchosen} + d(t)$ if the chosen option is the target (an Along trial) and $u_{chosen} - u_{unchosen} - d(t)$ if the chosen option is the competitor (an Against trial). Hence,

$$RT = \frac{T}{u_{chosen} - u_{unchosen} + (I)d(t)}, \quad (1)$$

where RT is the response time of trial t , $T > 0$ is the decision bound, $u_{chosen} - u_{unchosen}$ is the intrinsic utility difference between the chosen option and the undominated unchosen option of this trial and $d(t)$ is the strength of the decoy effect of this trial. The indicator function is $I = +1$ for an Along trial and $I = -1$ for an Against trial. The sign in front of $d(t)$ is different in Along and

560 Against trials, because the decoy effect favors a different option in the two cases.
 561 Assuming $d(t)$ is drawn from a normal distribution with subject-specific mean μ
 562 and standard deviation σ , we used maximum likelihood to estimate T , μ and σ
 563 for each subject. The trial-by-trial strength of the decoy effect $d(t)$ was backed
 564 out from (1). The LBA model uses the intrinsic utility difference and whether
 565 the decoy is helping the chosen option to estimate the RT and hence to back
 566 out $d(t)$.

567 The parameters estimated using (1) have the expected properties, suggesting
 568 the model is appropriate (Table 5). In particular, a subject's estimated average
 569 strength of the decoy effect, μ , is a measure of how "decoyable" she is. If the
 570 decoy makes the target more valuable, μ is expected to be positive. For all
 571 but two subjects, it is indeed positive. Furthermore, μ is strongly positively
 572 correlated with the number of preference reversals – the natural measure of how
 573 strongly decoys influence choices – suggesting that μ effectively captures how
 574 decoyable each subject is. As mentioned above, μ is also strongly positively
 575 correlated with the differential striatum activity in Along versus Against trials.
 576 This is evidence in favor of a link between striatal activity and the behavioral
 577 decoy effect. Since T can be interpreted as the threshold for making a decision,
 578 a larger T should imply a longer RT, and indeed the positive correlation between
 579 T and the average RT of decoy trials is very strong.

580 Neural Correlate of the Trial-by-trial Strength of the Decoy Effect

581 For each subject, we divided the decoy trials into two halves depending
 582 on the size of $d(t)$. The half with large $d(t)$ was classified as Strong trials,
 583 suggesting a strong decoy effect. The half with small $d(t)$ was classified as
 584 Weak trials. Neurally, no region was found contrasting Strong to Weak. The
 585 left IPL was the only region more active contrasting Weak to Strong (Figure 3A,
 586 Table 6). Its activity in fact parametrically tracks $d(t)$ negatively (Figure 3B,
 587 Table 7). The overlapping area is shown in Figure 3C. IPL has previously been
 588 implicated in goal-directed preparation of attention (Hopfinger et al., 2000; Raz

and Buhle, 2006; Shomstein, 2012). In particular, even though IPL is a large region, the area close to our activation cluster is activated more when there is a switch than when a task is repeated, when trials are incongruent than when they are congruent in the Stroop or flanker tasks, when stimulus and response are incompatible than when they are compatible, and when working memory is more heavily taxed (Sylvester et al., 2003; Liston et al., 2006a; McNab et al., 2008; Ye and Zhou, 2009). The common denominator in these experiments is that IPL may support the allocation of attention to facilitate task-relevant representations.

The fact that the target dominates the decoy is a salient aspect of the choice menu which may attract attention. This attention to a salient but irrelevant aspect may underlie the decoy effect (Krajbich et al., 2010). Switching attention to the relevant aspects of the menu such as the intrinsic properties of the undominated options would reduce the decoy effect. As IPL correlates negatively with $d(t)$, it is possible that attending less to decoys (as reflected by the negative correlation), subjects attend more to relevant information. This may be part of a control mechanism which corrects behavioral biases induced by decoys. However, unlike conventional control circuits, this mechanism would not necessarily slow down decision-making. When the competitor is intrinsically better than the target, the decoy effect works against choosing the competitor. If heightened attention means focusing on the intrinsic utilities of the undominated options, then choosing the competitor becomes easier and the decision may be faster. Behavior will become more rational and the RT will more closely reflect how long it would take to make a decision based only on the relevant aspects of the task.

Because the results of the Along-Against contrast suggest that context-dependent values could be manifested in differential striatal activity, we explored how IPL might differentially connect with the striatum depending on whether the decoy effect is successfully overridden. We did not find a direct modulation from IPL in our data. We did find a plausible indirect modulation through

619 rIFG, which we further verified by the TMS experiment. The indirect modula-
 620 tion is based on the following two PPI analyses. Since IPL plays a key role, we
 621 ran several robustness checks on its negative modulation of $d(t)$ as explained in
 622 (B). To summarize, in either a minor modification of (1) or in two attempts to
 623 measure the trial-by-trial strength of the decoy effect in a model-free way, IPL
 624 is significant at $p < 0.001$ uncorrected and has cluster size greater than 40 voxels
 625 (data not shown).

626 How Control Might Be Recruited: Two Connectivity Analyses

627 If the decoy effect increases the valuation of the target, it could potentially
 628 lead to the wrong decision if the target is not the intrinsically best option.
 629 Presumably, it is impossible to inhibit processing of the decoy at the perceptual
 630 level. On a higher level, domination may be a salient property that attracts
 631 attention, but control may mitigate the interference from the decoy and refocus
 632 attention on the intrinsic properties of the undominated options.

633 First Connectivity Analysis: between IPL and IFG

634 We divided the decoy trials into two halves. The half where the target has a
 635 lower intrinsic utility than the competitor is categorized as Conflict. The other
 636 half, where the target has the highest intrinsic utility, is categorized as NoCon-
 637 flict. In a Conflict trial, control may be required to override the decoy effect
 638 and prevent the intrinsically inferior target from being chosen. In a NoConflict
 639 trial, since the target is the intrinsically best option, the decoy effect is less
 640 problematic and control presumably less critical. We hypothesized a stronger
 641 connectivity between the area supporting attention and that implementing con-
 642 trol when attention is heightened and control is more likely to make a difference.

643 We took the average activity of a 4-mm sphere surrounding a peak voxel of
 644 IPL as the seed to examine whole-brain whether any area exhibits stronger func-
 645 tional connectivity with IPL in Conflict than in NoConflict trials as explained in

646 PPI 1. The rIFG is the only region that has this stronger task-related connectiv-
 647 ity (Figure 4A, Table 8). The rIFG is postulated to be a site where goal-directed
 648 and stimulus-driven attention converge (Asplund et al., 2010; Shomstein, 2012).
 649 Neuroimaging studies implicate this area in supporting inhibition to implement
 650 control (Dillon and Pizzagalli, 2007; Levy and Wagner, 2011; Nubert et al.,
 651 2012; Aron et al., 2014). It is more active in No-Go/Stop-Signal trials than in
 652 Go trials, in invalid Posner cueing trials than in valid ones, in trials where recent
 653 history may interfere than when it may not – all possibly reflecting processes
 654 related to overriding prepotent responses (Bunge et al., 2001; Levy and Wagner,
 655 2011). The rIFG has been implicated in a number of other exertions requiring
 656 self-control, such as inhibiting an incorrect answer to a problem of logic, fo-
 657 cusing on the facts rather than the framing of a question, or the decision to
 658 quit smoking (Goel and Dolan, 2003; De Martino et al., 2006; Berkman et al.,
 659 2011). Inhibiting the decoy effect likely is more critical to decision-making in a
 660 Conflict trial than in a NoConflict trial. The stronger functional connectivity
 661 between IPL and rIFG could possibly be due to the posterior area signaling
 662 the conflicting representations to the frontal cortex for control (Sylvester et al.,
 663 2003). Anatomical connections between IPL and IFG have been demonstrated
 664 using diffusion-weighted imaging (Rushworth et al., 2006; Caspers et al., 2011).
 665 As previous studies indicate that rIFG may play a role in inhibiting irrelevant
 666 responses, our connectivity result prompted us to explore further the role of
 667 rIFG in our data.

668 Second Connectivity Analysis: between IFG and Striatum

669 That the striatum is more active in Along than in Against trials is consistent
 670 with the hypothesis that the value of the same chosen option is higher when
 671 it is the target. This may lead to biased choices in Conflict trials, where by
 672 definition the target is intrinsically worse than the competitor. In Conflict trials,
 673 we say that control is unsuccessful if choice is along the decoy and successful if
 674 choice is against the decoy. Thus, by definition, if the intrinsically worse target
 675 was chosen, then control was unsuccessful; if the intrinsically better competitor

676 was chosen then control was successful. We hypothesized that when control is
 677 heightened, the connectivity between control and valuation areas will be stronger
 678 when control is successful than when control is unsuccessful.

679 We took the average activity of a 4-mm sphere surrounding a peak voxel
 680 of rIFG identified above as the seed to examine whether the striatum has a
 681 differential functional connectivity with rIFG when control was successful than
 682 when control was unsuccessful as explained in PPI 2. In a whole-brain search,
 683 the left striatum shows a stronger task-related functional connectivity with rIFG
 684 (Figure 4B, Table 9). The active cluster overlaps with what we discovered
 685 when contrasting Along to Against trials (Figure 4C). This is consistent with a
 686 possible role of rIFG in reducing the decoy-induced valuation bias. Anatomical
 687 connections between IFG and striatum have been demonstrated. Restricted
 688 frontostriatal diffusion seems to correlate with greater control, hinting at its
 689 contribution to the recruitment of control (Liston et al., 2006b).

690 The TMS Experiment: Temporarily Disrupting the rIFG

691 The two connectivity results lead us to hypothesize about the role of the rIFG
 692 in our experiment. Its differential functional connectivity with the striatum
 693 hints at its possible influence on choices. As it is often implicated in overriding
 694 irrelevant responses, rIFG may play a role in implementing control to overcome a
 695 potential decoy-induced decision-bias. We used TMS to investigate this further.
 696 We applied theta burst TMS to temporarily interfere with the region of interest
 697 before subjects started the choice task (Huang et al., 2005; Chao et al., 2011).
 698 In one group of subjects, the IFG group, the site of stimulation was at the peak
 699 voxel of rIFG identified from the fMRI experiment. In the other group, the
 700 vertex group, the site of stimulation was the vertex. Each group had thirty-two
 701 subjects, the same number as in the fMRI experiment. If rIFG plays a role in
 702 inhibiting the decoy-induced bias, then when it is temporarily disrupted, the
 703 inhibitory control is expected to be weaker, and the decoy effect is expected
 704 to be stronger (Jacobson et al., 2011). In terms of choices, we thus expected

the IFG group to exhibit more preference reversals than the vertex group. In terms of RT, because a strong decoy effect is expected to make it much easier to choose along the decoy than against it, we expected the IFG group to have a larger percentage decrease of RT in Along trials, compared with Against trials, than the vertex group.

These expectations were born out in the TMS experiment. The IFG group on average had a preference reversal rate of 25.30% whereas the vertex group had 18.72%, an increase of 6.58 percentage points. The t-test for comparing the preference reversal rate of the IFG group with that of the vertex group has a p-value (one-tailed) of 0.048 (Figure 5A). In behaviorally rational pairs, for the IFG group, on average, the RT was 7.23 seconds in Along trials and 9.08 seconds in Against trials. The average percentage decrease of RT in Along trials, compared with Against trials, was 19.22%. For the vertex group, on average, the RT was 6.68 seconds in Along trials and 7.47 seconds in Against behaviorally rational trials. The average percentage decrease of RT in Along trials, compared with Against trials, was 9.05%. The t-test for comparing the average percentage decrease of RT in Along trials, compared with Against trials, of the IFG group with that of the vertex group has a p-value (one-tailed) of 0.023 (Figure 5B).

4 Discussion

We have investigated the neural underpinning of the decoy effect. The fMRI study revealed differential striatal activity in Along versus Against trials. This is consistent with context-dependent valuations: the same chosen option seems more valuable when it is the target (even when there is no preference reversal). This differential activity correlates with two behavioral measures of the decoy effect, rendering it plausible that context-dependent valuation underlies the behavioral decoy effect. Our results do not contradict the most basic assumption of the standard rational choice model: subjects assign values to the feasible

options and choose the option with the highest value. However, if these values are context-dependent, cognitive control may be required to avoid choosing the target option when it is intrinsically worse. We found that IPL, often implicated in subserving voluntary attention, correlates negatively with the trial-by-trial strength of the decoy effect. IPL's functional connectivity with rIFG, a frontal area supporting inhibition to implement control, is stronger when control is more important. rIFG in turn exhibits stronger functional connectivity with striatum when control is successful. These results are summarized in Figure 6. Using TMS to temporarily disrupt rIFG to weaken control, subjects were found to exhibit a stronger decoy effect on behavior. Our results suggest a possible mechanism for how control is recruited and how it may influence valuations. Previous literature supports the role of frontal regions in control. Specifically, rIFG is often implicated in psychological tasks requiring the inhibition of prepotent irrelevant responses or implementation of self-control. The adjacent anterior prefrontal cortex was also found to be a key region in reducing a behavioral bias induced by an irrelevant rewarding feature (Scholl et al., 2015). Our results suggest that rIFG may be involved in inhibiting context-induced preference-reversals in simple choice tasks.

A possible source of context-dependence is divisive normalization, where the intrinsic utility of each option is divided by the total intrinsic utility of all feasible options (Louie et al., 2013). Thus, if A is more highly valued than B , then the positive normalized difference between A and B is smaller in $\{A, B, B'\}$ than in $\{A, B\}$, resulting in noisier choices and making B more likely to be chosen from the former menu. However, divisive normalization would also predict that B is more likely to be chosen from $\{A, B, A'\}$ than from $\{A, B\}$, contradicting what decoy experiments typically find. Thus, divisive normalization cannot explain the decoy effect. We compare choices along the decoy with choices against the decoy, thereby avoiding the possible confound of comparing three-option menus with two-option menus (Hedgcock and Rao, 2009). Suppose the better option A is chosen in a behaviorally rational pair. The normalized difference between

762 A and B is not systematically higher in $\{A, B, A'\}$ than in $\{A, B, B'\}$ because
 763 by design the total intrinsic utility of all options is not systematically lower
 764 in the former menu. Therefore divisive normalization provides no explanation
 765 for the systematic favoring of A in $\{A, B, A'\}$ and B in $\{A, B, B'\}$. Similarly,
 766 divisive normalization cannot explain the higher striatal activity in Along than
 767 in Against trials.

768 Range normalization might explain the systematic favoring of the target.
 769 Consider an $\{A, B, A'\}$ trial where A is more expensive than B . Then since A'
 770 is dominated by A , A' is even more expensive. Representing the very expensive
 771 decoy A' neurally may make the price differential between A and B less
 772 significant, hence increasing the attractiveness of A . The decoy effect may then
 773 be stronger, the more expensive is A' compared to A (Soltani et al., 2012).
 774 More generally, the distance between the target and the decoy might matter.
 775 Our experiment was not designed to investigate this, but our data does provide
 776 some support for the conjecture. We calculated the trial-by-trial intrinsic utility
 777 distance between the target and the decoy, $u_{target} - u_{decoy}$, using the logit
 778 estimates from (A). The average Pearson correlation coefficient between $u_{target} -$
 779 u_{decoy} and $d(t)$ is 0.42. Thus, within the range of variation of our experiment,
 780 the decoy effect seems stronger when the intrinsic utility distance is larger.
 781 However, a large part of the trial-by-trial variation in the decoy effect is not
 782 explained by the intrinsic utility distance between the target and the decoy.

783 Unavailable distractors have been found to affect value difference signals in
 784 ventral medial prefrontal cortex (vmPFC) (Chau et al., 2014). These distractors
 785 are close to the idea of decoys but did not have specificity, in the sense that the
 786 effect in principle was not specific to either the high-value or the low-value
 787 option. In contrast, the decoy is dominated by one option (the target) only, and
 788 thus has a strong specificity allowing us to explore how the value of an option
 789 depends on whether or not it is the target. Specifically, we found context-
 790 dependent activity in the striatum, suggesting a context-dependent valuation of
 791 the chosen option. The different results of the two studies are not surprising in

view of the different designs. Unlike distractors, decoys are possible objects of choice and may be considered as such at least initially, despite being dominated. Closer to decision, they may no longer be considered relevant options, but the effect of helping the target may linger. We find preliminary evidence (data not shown) supporting this conjecture. Modeling only the first two seconds of a trial, we find that the striatum likely codes the value sum of all options (including the decoy) but at that time, it is not more active in Along than in Against trials. In the middle two seconds, however, the striatum likely codes the value of the chosen option and no longer correlates with the value sum. Moreover, there is a striatal cluster of 173 voxels more active in Along than in Against trials. Even though fMRI does not have good time resolution to explore the dynamics, these observations are consistent with the hypothesis that decoys might initially be considered possible objects of choice, and later their effect is manifested as helping the targets. Among areas often correlated with values such as vMPFC and ACC, though we did not observe their activity in the first two seconds, we found it possibly correlating with the value of the chosen option in the middle two seconds. Hence the lack of their overall involvement in our experiment could be due to the long RTs. However, unlike striatum, they do not have differential activity in Along than in Against trials which is the main focus of our experiment. Various value-related signals seem distributed over the brain at different points of time. Biophysically realistic networks have been proposed (Wang, 2002; Hunt et al., 2014) which may produce various value-related signals. How these signals may be related to each other dynamically remains an open question.

The striatum is often implicated in conflict-eliciting choice situations, and the rIFG is involved in inhibiting a prepotent response. However, interpretations along this line do not easily apply to our experiment, where control may play a subtle role which does not simply show up in elevated mean activity. If striatal activity is linked to conflict-eliciting choices, we would expect it to be more active in Against trials than in Along trials (since choosing against the decoy

would presumably involve more of a conflict). In fact we observe the opposite. Moreover, our findings on IPL and rIFG suggest modulation of values rather than simple inhibition. In our data, IPL correlates with the $d(t)$ signal which is closely linked to values. It is difficult to interpret IPL activity as simple inhibition since, as can be seen from (1), $d(t)$ depends on the value difference and on which option is chosen. As for rIFG activity, it is often observed in No-Go or Stop-Signal tasks that manipulate control explicitly. Our experiment does not explicitly manipulate control: in each trial, subjects just choose an option. We find that the connectivity of rIFG with IPL is stronger in Conflict trials than in NoConflict trials and its connectivity with striatum is stronger when control is successful than when control is unsuccessful. Thus, intrinsic utility implicitly plays a role in our findings on rIFG as well. IPL is not known as a control region in the same way as rIFG is, but it often activates when attention is needed and in decision tasks (Nieuwenhuis et al., 2005). In view of this, a possible interpretation of our results is in terms of attention being important for the successful implementation of control.

In measuring the trial-by-trial strength of the decoy effect $d(t)$, we attributed variations in the strength of evidence favoring the chosen option to variations in $d(t)$. An alternative assumption would attribute them to random evaluation of intrinsic utility. However, note that when using (1) to estimate variations in $d(t)$, we take into account which option is target and which is decoy. This information would not be used to estimate a random intrinsic utility noise favoring one option over another. Hence the brain correlate of $d(t)$ is different from that of random intrinsic utility evaluations. Preliminary evidence (data not shown) shows that the IPL cluster which correlates negatively with $d(t)$ does not correlate with random utility noises. Among areas correlating with random utility noises, most noticeable is the vmPFC which often codes values.

In our experiment, the decoy is irrelevant to the choice problem – it would be rational to simply disregard the decoy and let the brain save its limited firing range for the viable (undominated) options. However, a comparative valuation

852 heuristic could be adaptive in naturally occurring sequential-choice situations,
 853 where whether an organism should choose an option right away depends on the
 854 background (Houston, 1997). This heuristic requires the use of memories and
 855 contextual information. In an experiment with European starlings, the birds
 856 rely on information from the context to make the optimal choice in sequen-
 857 tial encounters (Freidin and Kacelnik, 2011). A comparative valuation heuristic
 858 may be evolutionarily conserved because sequential decision problems have been
 859 prevalent. However, there are several reasons why the heuristic may lead to sub-
 860 optimal decisions in modern-day choice situations (Slovic et al., 2007). First,
 861 someone may be trying to manipulate the decision maker. For example, a prod-
 862 uct’s intrinsic properties of price and quality will determine how well it will
 863 satisfy the consumer’s wants and needs, i.e., will determine the experienced (or
 864 intrinsic) utility – but by introducing artificial decoy options, marketers distort
 865 the decision utility (i.e., the consumer’s assessment of the intrinsic utility). *The*
 866 *Economist*’s advertisement mentioned above may be an example of this. Sec-
 867 ond, the assumptions underlying the normal operation of the system may be
 868 violated. For example, the heuristics may be well adapted to naturally occur-
 869 ring sequential decision problems, but not to the static choice problems in our
 870 experiment. Third, some decision problems are too complex for purely affec-
 871 tive judgments and require more deliberative and “rational” cognitive processes.
 872 All three reasons are relevant to our experiment. Rather than being a “design
 873 flaw,” the decoy effect may be an evolved adaptation that illuminates the on-
 874 line comparative judgments of the decision-making system. It may be analogous
 875 to visual illusions, e.g., the Ebbinghaus illusion where the perceived size of an
 876 object depends comparatively on the sizes of neighboring objects (Goodale and
 877 Haffenden, 1998; Eagleman, 2001), which is adaptive in ecologically relevant sce-
 878 narios. Similarly, the heuristic paying attention to dominance may have been
 879 adaptive in the past, but modern-day shopping situations may require control,
 880 which could have its function originating elsewhere, to prevent biased decisions.

References

- Ariely D (2008) Predictably irrational : the hidden forces that shape our decisions, 1st Edition. New York, NY: Harper.
- Aron AR, Robbins TW, Poldrack RA (2014) Inhibition and the right inferior frontal cortex: one decade on. *Trends in cognitive sciences* 18:177-185.
- Asplund CL, Todd JJ, Snyder AP, Marois R (2010) A central role for the lateral prefrontal cortex in goal-directed and stimulus-driven attention. *Nature neuroscience* 13:507-512.
- Bateson M, Healy SD, Hurly TA (2003) Context-dependent foraging decisions in rufous hummingbirds. *P Roy Soc B-Biol Sci* 270:1271-1276.
- Berkman ET, Falk EB, Lieberman MD (2011) In the trenches of real-world self-control: neural correlates of breaking the link between craving and smoking. *Psychological science* 22:498-506.
- Brown SD, Heathcote A (2008) The simplest complete model of choice response time: linear ballistic accumulation. *Cognitive psychology* 57:153-178.
- Bunge SA, Ochsner KN, Desmond JE, Glover GH, Gabrieli JD (2001) Prefrontal regions involved in keeping information in and out of mind. *Brain : a journal of neurology* 124:2074-2086.
- Caspers S, Eickhoff SB, Rick T, von Kapri A, Kuhlen T, Huang R, Shah NJ, Zilles K (2011) Probabilistic fibre tract analysis of cytoarchitectonically defined human inferior parietal lobule areas reveals similarities to macaques. *NeuroImage* 58:362-380.
- Chao CM, Tseng P, Hsu TY, Su JH, Tzeng OJ, Hung DL, Muggleton NG, Juan CH (2011) Predictability of saccadic behaviors is modified by transcranial magnetic stimulation over human posterior parietal cortex. *Human brain mapping* 32:1961-1972.
- Chau BK, Kolling N, Hunt LT, Walton ME, Rushworth MF (2014) A neural mechanism underlying failure of optimal choice with multiple options. *Nature neuroscience* 17:463-470.
- De Martino B, Kumaran D, Seymour B, Dolan RJ (2006) Frames, biases, and rational decision-making in the human brain. *Science* 313:684-687.

- 912 Dillon DG, Pizzagalli DA (2007) Inhibition of Action, Thought, and Emo-
 913 tion: A Selective Neurobiological Review. *Applied & preventive psychology*
 914 : journal of the American Association of Applied and Preventive Psychology
 915 12:99-114.
- 916 Eagleman DM (2001) Visual illusions and neurobiology. *Nature reviews*
 917 Neuroscience 2:920-926.
- 918 Eklund A, Nichols TE, Knutsson H (2016) Cluster failure: Why fMRI in-
 919 ferences for spatial extent have inflated false-positive rates. *Proceedings of the*
 920 National Academy of Sciences of the United States of America 113:7900-7905.
- 921 Freidin E, Kacelnik A (2011) Rational choice, context dependence, and the
 922 value of information in European starlings (*Sturnus vulgaris*). *Science* 334:1000-
 923 1002.
- 924 Goel V, Dolan RJ (2003) Explaining modulation of reasoning by belief. *Cog-*
 925 nition 87:B11-22.
- 926 Goodale MA, Haffenden A (1998) Frames of reference for perception and
 927 action in the human visual system. *Neuroscience and biobehavioral reviews*
 928 22:161-172.
- 929 Hedgcock W, Rao AR (2009) Trade-Off Aversion as an Explanation for the
 930 Attraction Effect: A Functional Magnetic Resonance Imaging Study. *J Market-*
 931 ing Res 46:1-13.
- 932 Hopfinger JB, Buonocore MH, Mangun GR (2000) The neural mechanisms
 933 of top-down attentional control. *Nature neuroscience* 3:284-291.
- 934 Houston AI (1997) Natural selection and context-dependent values. *P Roy*
 935 Soc B-Biol Sci 264:1539-1541.
- 936 Huang YZ, Edwards MJ, Rounis E, Bhatia KP, Rothwell JC (2005) Theta
 937 burst stimulation of the human motor cortex. *Neuron* 45:201-206.
- 938 Huber J, Payne JW, Puto C (1982) Adding Asymmetrically Dominated
 939 options - Violations of Regularity and the Similarity Hypothesis. *J Consum*
 940 Res 9:90-98.
- 941 Hunt LT, Dolan RJ, Behrens TE (2014) Hierarchical competitions subserving
 942 multi-attribute choice. *Nature neuroscience* 17:1613-1622.

- Jacobson L, Javitt DC, Lavidor M (2011) Activation of inhibition: diminishing impulsive behavior by direct current stimulation over the inferior frontal gyrus. *Journal of cognitive neuroscience* 23:3380-3387.
- Kim JG, Biederman I, Juan CH (2011) The benefit of object interactions arises in the lateral occipital cortex independent of attentional modulation from the intraparietal sulcus: a transcranial magnetic stimulation study. *The Journal of neuroscience : the official journal of the Society for Neuroscience* 31:8320-8324.
- Krajibich I, Armel C, Rangel A (2010) Visual fixations and the computation and comparison of value in simple choice. *Nature neuroscience* 13:1292-1298.
- Latty T, Beekman M (2011) Irrational decision-making in an amoeboid organism: transitivity and context-dependent preferences. *P Roy Soc B-Biol Sci* 278:307-312.
- Levy BJ, Wagner AD (2011) Cognitive control and right ventrolateral prefrontal cortex: reflexive reorienting, motor inhibition, and action updating. *Annals of the New York Academy of Sciences* 1224:40-62.
- Liston C, Matalon S, Hare TA, Davidson MC, Casey BJ (2006a) Anterior cingulate and posterior parietal cortices are sensitive to dissociable forms of conflict in a task-switching paradigm. *Neuron* 50:643-653.
- Liston C, Watts R, Tottenham N, Davidson MC, Niogi S, Ulug AM, Casey BJ (2006b) Frontostriatal microstructure modulates efficient recruitment of cognitive control. *Cerebral cortex* 16:553-560.
- Louie K, Khaw MW, Glimcher PW (2013) Normalization is a general neural mechanism for context-dependent decision making. *Proceedings of the National Academy of Sciences of the United States of America* 110:6139-6144.
- Manes JL, Parkinson AL, Larson CR, Greenlee JD, Eickhoff SB, Corcos DM, Robin DA (2014) Connectivity of the subthalamic nucleus and globus pallidus pars interna to regions within the speech network: a meta-analytic connectivity study. *Human brain mapping* 35:3499-3516.
- McNab F, Leroux G, Strand F, Thorell L, Bergman S, Klingberg T (2008) Common and unique components of inhibition and working memory: an fMRI,

- 974 within-subjects investigation. *Neuropsychologia* 46:2668-2682.
- 975 Muggleton NG, Cowey A, Walsh V (2008) The role of the angular gyrus
976 in visual conjunction search investigated using signal detection analysis and
977 transcranial magnetic stimulation. *Neuropsychologia* 46:2198-2202.
- 978 Nieuwenhuis S, Heslenfeld DJ, von Geusau NJ, Mars RB, Holroyd CB, Yeung
979 N (2005) Activity in human reward-sensitive brain areas is strongly context
980 dependent. *NeuroImage* 25:1302-1309.
- 981 Nubert FX, Mars RB, Rushworth MFS (2012) Is there an inferior frontal
982 cortical network for cognitive control and inhibition? In: *Principles of frontal*
983 *lobe function* (Stuss DT, Knight RT, eds). pp332-352. New York: Oxford UP.
- 984 Padoa-Schioppa C, Assad JA (2006) Neurons in the orbitofrontal cortex
985 encode economic value. *Nature* 441:223-226.
- 986 Padoa-Schioppa C, Assad JA (2008) The representation of economic value
987 in the orbitofrontal cortex is invariant for changes of menu. *Nature neuroscience*
988 11:95-102.
- 989 Raz A, Buhle J (2006) Typologies of attentional networks. *Nature reviews*
990 *Neuroscience* 7:367-379.
- 991 Rushworth MF, Behrens TE, Johansen-Berg H (2006) Connection patterns
992 distinguish 3 regions of human parietal cortex. *Cerebral cortex* 16:1418-1430.
- 993 Sasaki T, Pratt SC (2011) Emergence of group rationality from irrational
994 individuals. *Behav Ecol* 22:276-281.
- 995 Scholl J, Kolling N, Nelissen N, Wittmann MK, Harmer CJ, Rushworth MF
996 (2015) The Good, the Bad, and the Irrelevant: Neural Mechanisms of Learning
997 Real and Hypothetical Rewards and Effort. *The Journal of neuroscience : the*
998 *official journal of the Society for Neuroscience* 35:11233-11251.
- 999 Shafir S, Waite TA, Smith BH (2002) Context-dependent violations of ratio-
1000 nal choice in honeybees (*Apis mellifera*) and gray jays (*Perisoreus canadensis*).
1001 *Behav Ecol Sociobiol* 51:180-187.
- 1002 Shomstein S (2012) Cognitive functions of the posterior parietal cortex: top-
1003 down and bottom-up attentional control. *Frontiers in integrative neuroscience*
1004 6:38.

- 1005 Slovic P, Finucane ML, Peters E, MacGregor DG (2007) The affect heuristic.
1006 Eur J Oper Res 177:1333-1352.
- 1007 Soltani A, De Martino B, Camerer C (2012) A range-normalization model
1008 of context-dependent choice: a new model and evidence. PLoS computational
1009 biology 8:e1002607.
- 1010 Stewart LM, Walsh V, Rothwell JC (2001) Motor and phosphene thresholds:
1011 a transcranial magnetic stimulation correlation study. Neuropsychologia 39:415-
1012 419.
- 1013 Sylvester CY, Wager TD, Lacey SC, Hernandez L, Nichols TE, Smith EE,
1014 Jonides J (2003) Switching attention and resolving interference: fMRI measures
1015 of executive functions. Neuropsychologia 41:357-370.
- 1016 Wang XJ (2002) Probabilistic decision making by slow reverberation in cor-
1017 tical circuits. Neuron 36:955-968.
- 1018 Ye Z, Zhou X (2009) Conflict control during sentence comprehension: fMRI
1019 evidence. NeuroImage 48:280-290.

Figure Legends

Figure 1A: Sample screens from the experiment. Subjects were endowed with 1500 New Taiwan Dollars to spend in each trial and made choices from a series of two- and three-option menus. The options on the menus were restaurant meals with specified prices and quality levels in which higher percentages imply better qualities. The trial in the top left corner is a two-option trial whereas that in the top right corner is a three-option trial. The two trials in the bottom panel are a matching pair of decoy trials. In this example, option *A* has quality 72% and price \$680; option *B* has quality 95% and price \$1010; option *E* has quality 84% and price \$900; option *A'* has quality 61% and price \$690; option *B'* has quality 94% and price \$1110. An option is dominated by another option if it has both lower quality and higher price. Thus, *A'* is dominated by *A* but not by *B*, whereas *B'* is dominated by *B* but not by *A*. The target, denoted *T*, is the option that dominates the decoy, denoted *D*; there is no domination relationship between the competitor, *C*, and the decoy. Hence in the bottom left trial, the target is *A* (quality 72% and price \$680), the decoy is *A'* (61%, \$690) and the competitor is *B* (95%, \$1010). In the bottom right trial, *B* (95%, \$1010) is the target, *B'* (94%, \$1110) is the decoy and *A* (72%, \$680) is the competitor.

Figure 1B: Percentage of preference reversals and anti choices. The figure is a subject-by-subject breakdown of the percentage of preference reversals and anti choices of decoy trials. We number subjects in descending order of the former. Preference reversals occur when *A* is chosen from $\{A, B, A'\}$ and *B* from $\{A, B, B'\}$ whereas anti choices occur when *B* is chosen from $\{A, B, A'\}$ and *A* from $\{A, B, B'\}$. Two subjects had neither preference reversals nor anti choices.

Figure 1C Shorter response times in Along than in Against behaviorally rational pairs. The figure is a subject-by-subject breakdown of the percentage decrease of RT in Along than in Against behaviorally rational pairs. We number

1049 subjects in descending order. The percentage decrease of RT is the difference
 1050 of RT in Against trials and Along trials divided by the average of them in
 1051 percentage.

1052 Figure 2A The left striatum (shown at $y=8$) correlated parametrically with
 1053 the intrinsic utility of the chosen option positively [$P<0.05$ at whole-brain cluster
 1054 correction].

1055 Figure 2B The left striatum was more active in Along than in Against trials
 1056 (shown at $y=8$) [$P<0.05$ at whole-brain cluster correction].

1057 Figure 2C Overlap. The left striatum was more active in Along than in
 1058 Against trials (shown at $y=8$) (shown in yellow at $P<0.001$, uncorrected) and
 1059 correlated parametrically with the intrinsic utility of the chosen option positively
 1060 (shown in red at $P<0.001$, uncorrected). The cluster of the left striatum is
 1061 circled and the overlapping area is shown in orange.

1062 Figure 2D Positive correlation between the number of preference reversals
 1063 and the differential striatal Along versus Against activity. The robust regression
 1064 slope is 0.0064 and the p-value (two-tailed) is 0.065. The putamen activity is
 1065 based on a 4-mm sphere centered at $(-20, 12, 2)$.

1066 Figure 2E Positive correlation between the average strength of the decoy
 1067 effect μ and the differential striatal Along versus Against activity. The robust
 1068 regression slope is 0.1833 and the p-value (two-tailed) is 0.009. The putamen
 1069 activity is based on a 4-mm sphere centered at $(-20, 12, 2)$.

1070 Figure 3A Binary Weak versus Strong decoy effect. The left IPL was more
 1071 active in Weak than in Strong trials (shown at $y=-40$) [$P<0.05$ at whole-brain
 1072 cluster correction].

1073 Figure 3B Parametric modulation from the LBA model. The left IPL para-
 1074 metrically correlated with $d(t)$ negatively (shown at $y=-40$) [$P<0.05$ at whole-
 1075 brain cluster correction].

1076 Figure 3C Overlap. The left IPL was more active in Weak trials than in
1077 Strong trials (shown at $y=-40$) (shown in yellow at $P<0.001$, uncorrected) and
1078 parametrically correlated with $d(t)$ negatively (shown in red at $P<0.001$, uncor-
1079 rected). The cluster of the left IPL is circled and the overlapping area is shown
1080 in orange.

1081 Figure 4A The rIFG exhibited stronger functional connectivity with the IPL
1082 in Conflict than in NoConflict trials (shown at $y=16$) [$P<0.05$ at whole-brain
1083 cluster correction]. The psychophysiological interaction is based on a 4-mm
1084 sphere of IPL centered at $(-40, -40, 46)$.

1085 Figure 4B The left striatum exhibited stronger functional connectivity with
1086 the rIFG in SuccessfulControl than in UnsuccessfulControl trials (shown at $y=8$)
1087 [$P<0.05$ at whole-brain cluster correction]. The psychophysiological interaction
1088 is based on a 4-mm sphere of rIFG centered at $(60, 16, 22)$.

1089 Figure 4C Overlap. The left striatum was more active in Along than in
1090 Against trials ($y=8$) (shown in yellow at $P<0.001$, uncorrected) and exhibited
1091 stronger functional connectivity with the rIFG in SuccessfulControl than in
1092 UnsuccessfulControl trials (shown in red at $P<0.001$, uncorrected). The cluster
1093 of the left striatum is circled and the overlapping area is shown in orange.

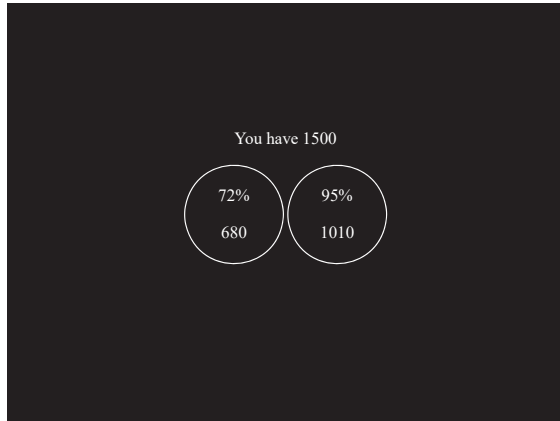
1094 Figure 5A Larger percentage of preference reversals in the IFG group than
1095 in the vertex group. Overlaid on the mean and the SEM are each subject's data
1096 point.

1097 Figure 5B Larger percentage decrease of response time in the IFG group
1098 than in the vertex group. The response times were shorter in Along than in
1099 Against behaviorally rational pairs and this was even more so in the IFG group
1100 than in the vertex group. Overlaid on the mean and the SEM are each subject's
1101 data point.

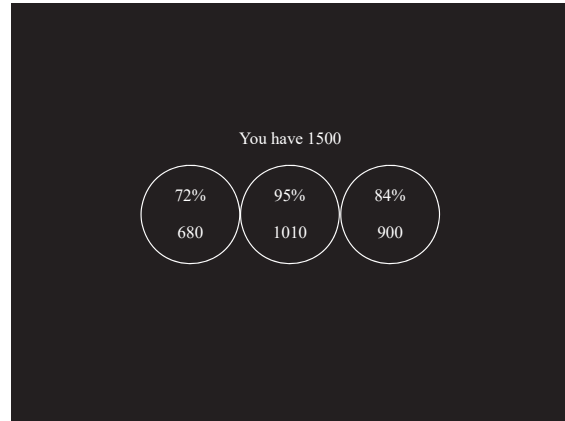
1102 Figure 6 An illustration of the main results.

Figure 1

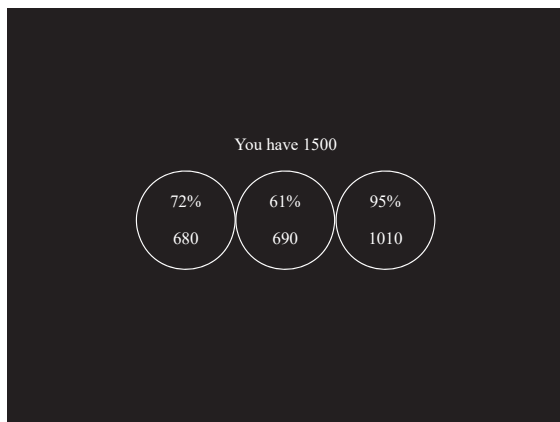
A



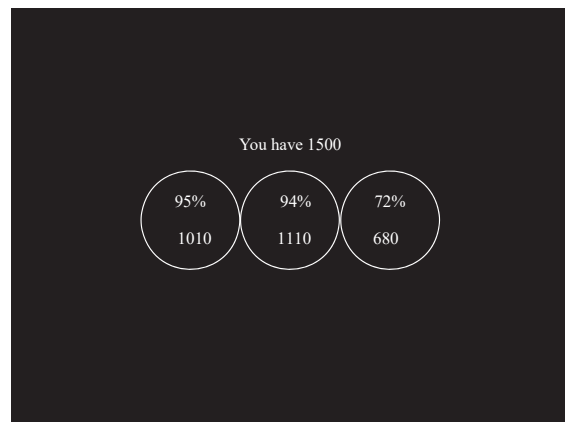
an {A, B} trial



an {A, B, E} trial

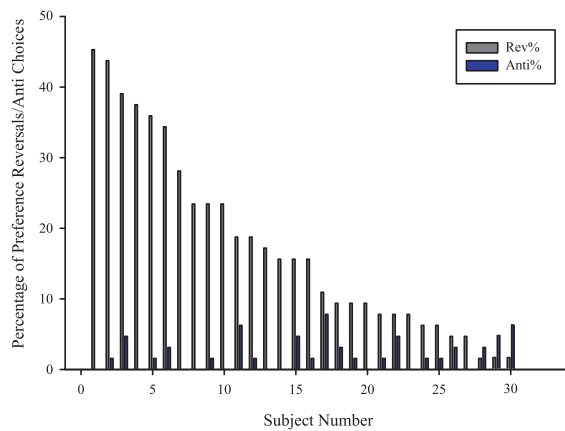


an {A, B, A'} trial in the order of AA'B or TDC



an {A, B, B'} trial in the order of BB'A or TDC

B



C

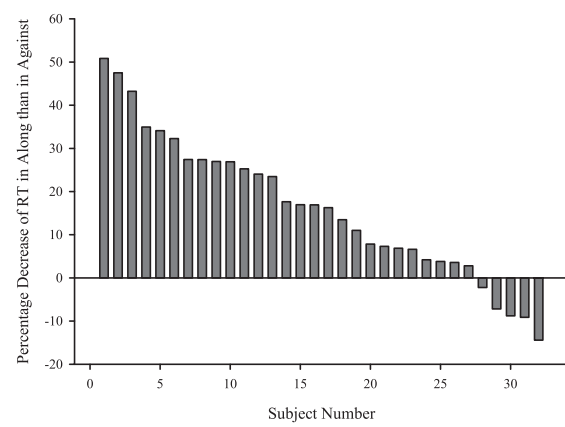
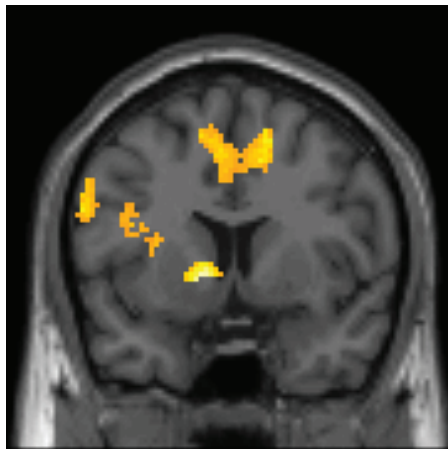


Figure 2

A Intrinsic Chosen Utility



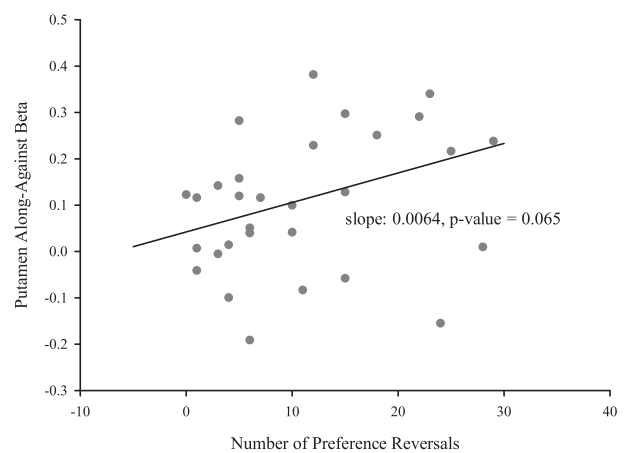
B Along versus Against



C Overlap



D



E

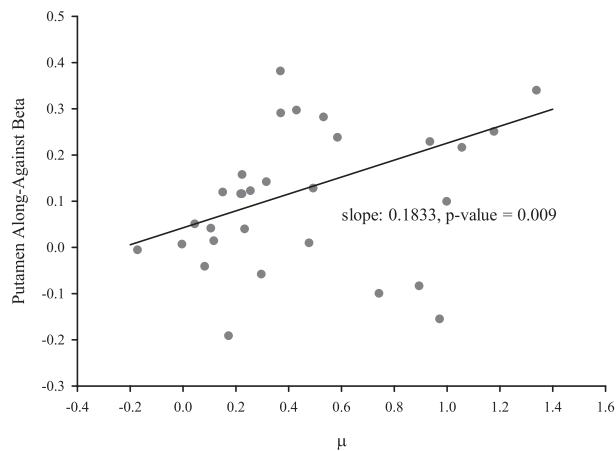
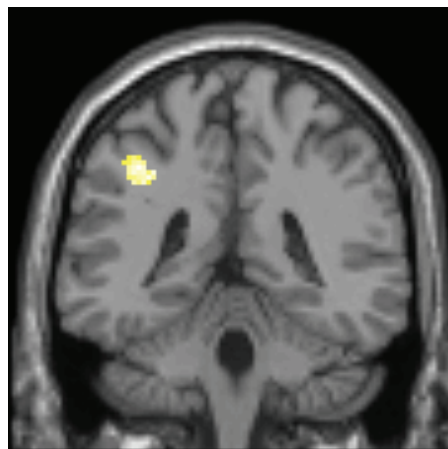


Figure 3

A Weak versus Strong



B Parametric



C Overlap

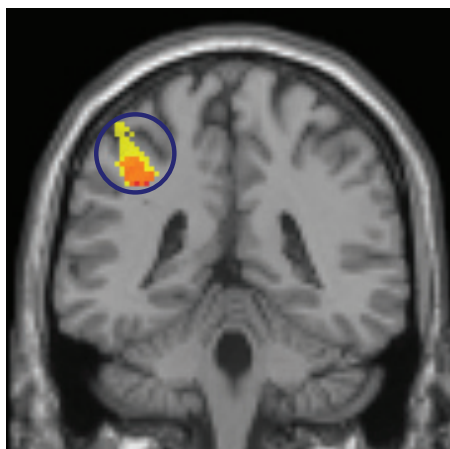


Figure 4

A Conflict versus NoConflict



B SuccessfulControl versus UnsuccessfulControl



C Overlap

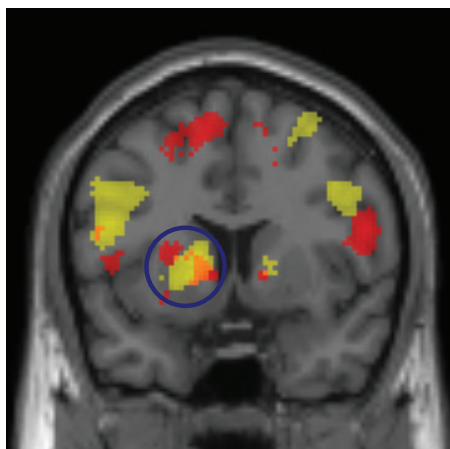
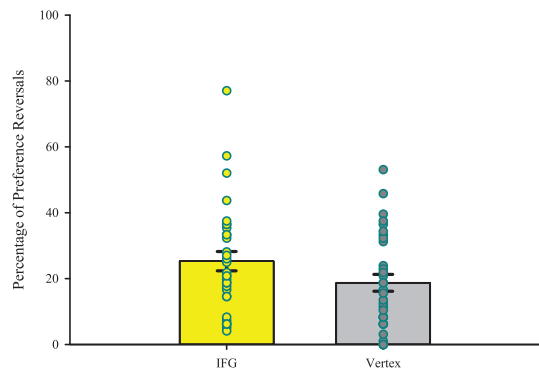


Figure 5

A



B

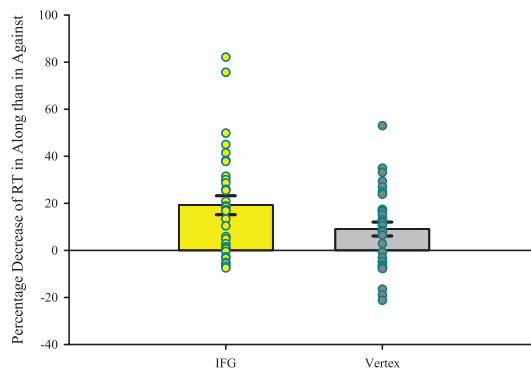


Figure 6

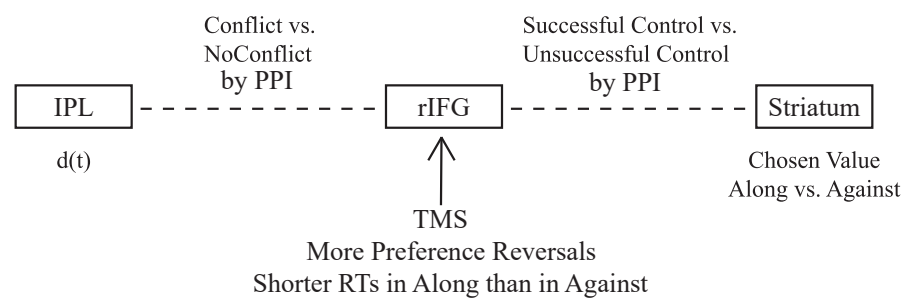


Table 1 Regions parametrically correlating with the utility of the chosen option positively (GLM 1)

Cluster Size	Brain Region	MNI Coordinate			Z Value
7324	Left Premotor and Striatum				
	BA 6	-58	2	38	5.18
	BA 9	-58	4	26	5.06
	BA 4	-36	-30	58	4.80
	BA 24	-2	0	46	4.40
	SMA	14	0	64	4.29
	Caudate	-12	10	0	4.79
	Putamen	-16	8	0	3.74
347	Medial Frontal Gyrus				
		-8	50	-14	4.03
985	Middle Frontal Gyrus				
	BA 9	48	2	34	4.35
	BA 6	58	2	42	4.27
337	Precentral Gyrus				
	BA 6	30	-14	56	3.95
323	Postcentral Gyrus				
	BA 2	50	-18	26	3.72
277	Middle Occipital gyrus				
	BA 19	52	-70	4	4.04
4157	Occipital Cortex				
	BA 19	-24	-78	24	4.32
	Cuneus	20	-84	24	4.30
	BA 18	-4	-68	2	4.09

(1) Local maxima of clusters with $p < 0.05$ at whole-brain cluster correction with a t threshold of 3.40 and an extent of 277 voxels are shown.

(2) There is no region parametrically correlating with the utility of the chosen option negatively.

Table 2 Regions more active in Along than in Against trials (GLM 1)

Cluster Size	Brain Region	MNI Coordinate			Z Value
1587	Left IFG				
	BA 46	-40	42	10	4.9
	BA 9	-50	6	22	4.4
	BA 10	-36	34	8	4.27
	BA 6	-30	-6	56	3.71
910	Right IFG				
	BA 46	40	32	20	4.17
	BA 10	38	40	12	4.16
	BA 9	42	4	34	4.11
549	Striatum/Insula				
	Insula	-28	18	4	4.79
	caudate	-12	8	8	4.22
	putamen	-20	8	0	3.86
530	Premotor Cortex				
	BA 6	22	-6	56	3.96
	BA 6	32	4	62	3.81
	BA 6	30	18	58	3.77
1356	Right IPL				
	BA 40	34	-60	42	5.42
	BA 40	40	-56	50	4.59
	precuneus	12	-70	50	4.39
917	Left IPL				
	BA 40	-30	-64	46	4.84
	BA 40	-38	-58	48	4.42
	BA 40	-44	-44	46	4.09

(1) Local maxima of clusters with $p < 0.05$ at whole-brain cluster correction with a t threshold of 3.40 and an extent of 530 voxels are shown.

(2) IFG stands for inferior frontal gyrus and IPL stands for inferior parietal lobule.

(3) There is no region more active in Against than in Along trials.

Table 3 Positive correlation between the number of preference reversals and the differential striatal activity Along versus Against in the top panel and positive correlation between μ and the differential striatal activity Along versus Against in the bottom panel.

		(-20, 12, 2)	(-22, 10, -2)	(-24, 14, -2)	(-12, 14, 4)
Reversals	Slope	0.0064	0.0076	0.0051	0.0045
	p-value	0.065*	0.025**	0.063*	0.141
μ	Slope	0.1833	0.1319	0.1047	0.2010
	p-value	0.009***	0.099*	0.093*	0.003***

(1) The striatum activity is based on a 4-mm sphere centered at putamen (-20, 12, 2), putamen (-22, 10, -2), putamen (-24, 14, -2) and caudate (-12, 4, 4) (from left to right).

(2) The slopes of the robust regressions and their associated p-values (two-tailed) are provided.

(3) * denotes significance at 10% level; ** at 5% level; *** at 1% level.

Table 4 Regions parametrically correlating with the intrinsic utility difference of the chosen option versus the undominated unchosen option positively (GLM 1)

Cluster Size	Brain Region	MNI Coordinate			Z Value
481	Premotor Cortex				
	BA 6	-58	0	38	4.7
2039	Sensorimotor Cortex				
	BA 4	-28	-16	60	4.1
	BA 3	-46	-20	62	4.0
	BA 6 (medial)	-4	2	42	3.9
888	Visual cortex				
	Cuneus	14	-84	18	3.7
	Cuneus	-12	-68	4	3.5
511	Inferior Frontal Gyrus				
	BA 9	48	2	34	4.2
	BA 6	58	2	44	4.0

(1) Local maxima of clusters with $p < 0.05$ at whole-brain cluster correction with a t threshold of 3.40 and an extent of 481 voxels are shown.

(2) There is no region parametrically correlating with the intrinsic utility difference of the chosen option versus the undominated unchosen option negatively.

Table 5 Strong positive correlation between the number of preference reversals and μ , strong positive correlation between the percentage decrease of RT in Along than in Against and μ , and strong positive correlation between the average RT in decoy trials and T (from left to right).

		Rev vs. μ	% vs. μ	RT vs. T
Pearson Correlation	Coefficient	0.610	0.517	0.589
	p-value	0.000***	0.002***	0.000***
Robust Regression	Slope	0.027	0.012	1.310
	p-value	0.001***	0.002***	0.001***

(1) Rev stands for the number of reversals; % the percentage decrease of RT in Along than in Against; RT the average RT in decoy trials.

(2) The Pearson correlation coefficients and their associated p-values (two-tailed) are provided in the upper panel. The slopes of the robust regressions and their associated p-values (two-tailed) are provided in the bottom panel.

(3) * denotes significance at 10% level; ** at 5% level; *** at 1% level.

Table 6 Regions more active in Weak than in Strong trials (GLM 2)

Cluster Size	Brain Region	MNI Coordinate			Z Value
461	Inferior Parietal Lobule				
	BA 40	-36	-42	44	4.1
	BA 40	-40	-40	46	4.1
	BA 40	-38	-38	42	4.1

(1) Local maxima of clusters with $p < 0.05$ at whole-brain cluster correction with a t threshold of 3.40 and an extent of 461 voxels are shown.

(2) There is no region more active in Strong than in Weak trials.

Table 7 Regions parametrically correlating with the trial-by-trial decoy effect $d(t)$ negatively (GLM 3)

Cluster Size	Brain Region	MNI Coordinate			Z Value
234	Medial Frontal Gyrus				
	BA 32	-16	46	-4	4.5
	BA 10	-12	54	4	4.0
185	Inferior Parietal Lobule				
	BA 40	-36	-42	44	3.9
	BA 40	-40	-40	46	3.4

(1) Local maxima of clusters with $p < 0.05$ at whole-brain cluster correction with a t threshold of 3.40 and an extent of 185 voxels are shown.

(2) There is no region parametrically correlating with the trial-by-trial decoy effect $d(t)$ positively.

Table 8 Regions exhibiting stronger functional connectivity with IPL in Conflict than in NoConflict trials (PPI 1)

Cluster Size	Brain Region	MNI Coordinate			Z Value
342	Inferior Frontal Gyrus				
	BA9	60	16	22	4.6

(1) Local maxima of clusters with $p < 0.05$ at whole-brain cluster correction with a t threshold of 3.40 and an extent of 342 voxels are shown. The psychophysiological interaction is based on a 4-mm sphere of IPL centered at (-40, -40, 46).

(2) There is no region exhibiting stronger functional connectivity with IPL in NoConflict than in Conflict trials.

Table 9 Regions exhibiting stronger functional connectivity with rIFG in SuccessfulControl than in UnsuccessfulControl trials (PPI 2)

Cluster Size	Brain Region	MNI Coordinate			Z Value
2085	Striatum				
	Medial Globus Pallidus	-10	2	0	5.0
	Caudate	-8	4	0	4.6
	Putamen	-22	0	4	4.0
660	Inferior Frontal Gyrus				
	BA 9	60	16	22	4.8
	BA 45	42	24	6	4.3
616	Precentral Gyrus				
	BA 6	-10	2	64	4.4
	BA 6	-36	-10	40	4.4
221	Medial Frontal Gyrus				
	BA 24	16	2	40	4.4
	BA 6	12	2	58	3.6
286	Postcentral Gyrus				
	BA 3	46	-22	46	4.0
	BA 3	38	-24	44	4.0
356	Inferior Frontal Gyrus				
	BA 44	-54	14	14	4.0
	BA 13	-36	22	2	3.7
188	Cerebellum				
		12	-60	-22	3.9
		-6	-52	-20	3.8

(1) Local maxima of clusters with $p < 0.05$ at whole-brain cluster correction with a t threshold of 3.41 and an extent of 188 voxels are shown. The psychophysiological interaction is based on a 4-mm sphere of IFG centered at (60, 16, 22).

(2) There is no region exhibiting stronger functional connectivity with IFG in UnsuccessfulControl than in SuccessfulControl trials.